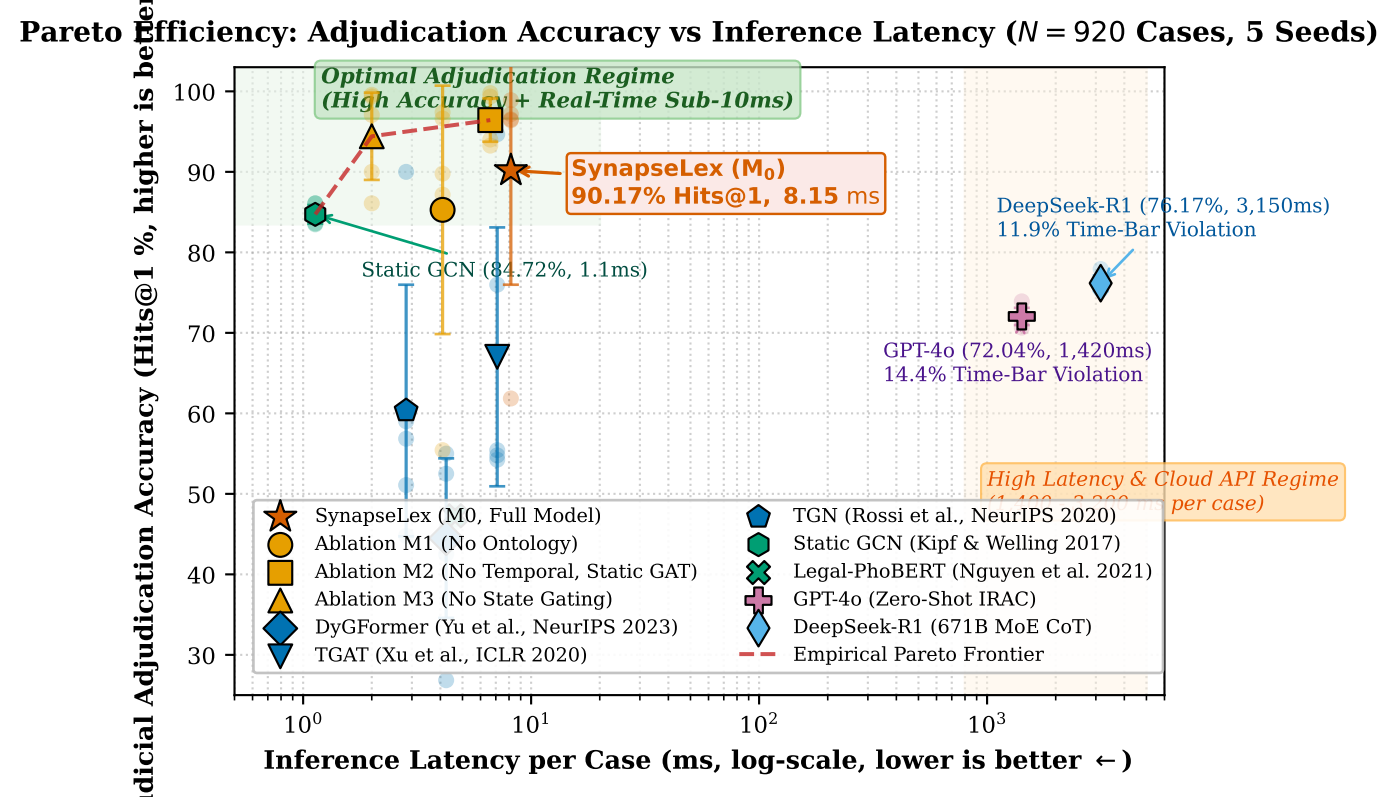

Pareto Efficiency: Adjudication Accuracy vs Inference Latency ($N = 920$ Cases, 5 Seeds)
Optimal Adjudication Regime
(High Accuracy + Real-Time Sub-10ms)
SynapseLex (M_0)
90.17% Hits@1, 8.15 ms
Static GCN (84.72%, 1.1ms)
DeepSeek-R1 (76.17%, 3,150ms)
11.9% Time-Bar Violation
GPT-4o (72.04%, 1,420ms)
14.4% Time-Bar Violation
High Latency & Cloud API Regime
per case)
SynapseLex (M0, Full Model)
Ablation M1 (No Ontology)
Ablation M2 (No Temporal, Static GAT)
Ablation M3 (No State Gating)
DyGFormer (Yu et al., NeurIPS 2023)
TGAT (Xu et al., ICLR 2020)
TGN (Rossi et al., NeurIPS 2020)
Static GCN (Kipf & Welling 2017)
Legal-PhoBERT (Nguyen et al. 2021)
GPT-4o (Zero-Shot IRAC)
DeepSeek-R1 (671B MoE CoT)
Empirical Pareto Frontier
Judicial Adjudication Accuracy (Hits@1 %, higher is better ↑)
Inference Latency per Case (ms, log-scale, lower is better ←)
100
90
80
70
60
50
40
30
10^0
10^1
10^2
10^3